One-Page Website Creation & Deployment using Replit

In Partial Fulfillment of the Requirements for:

GE 4120 Elective 3

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Initial Prompt

"Create a one-page technical reference website titled 'UAV Avionics & Wiring Guide.' The design should be 'Industrial-Tech' with a dark theme, neon blue accents, and a grid-pattern background. Include these sections:

Diagram Header: A hero section with a title 'Mastering UAV Power Distribution.'

Interactive Diagram: A centered, clean SVG or styled CSS box representing a flight controller (FC) connected to 4 ESCs, a LiPo battery, and a Radio Receiver (RX).

Component Breakdown: A list describing the wiring logic (e.g., Signal wires, XT60 power leads, and 5V/12V BECs).

Importance: A section on why correct wire gauge and soldering are critical for flight safety."

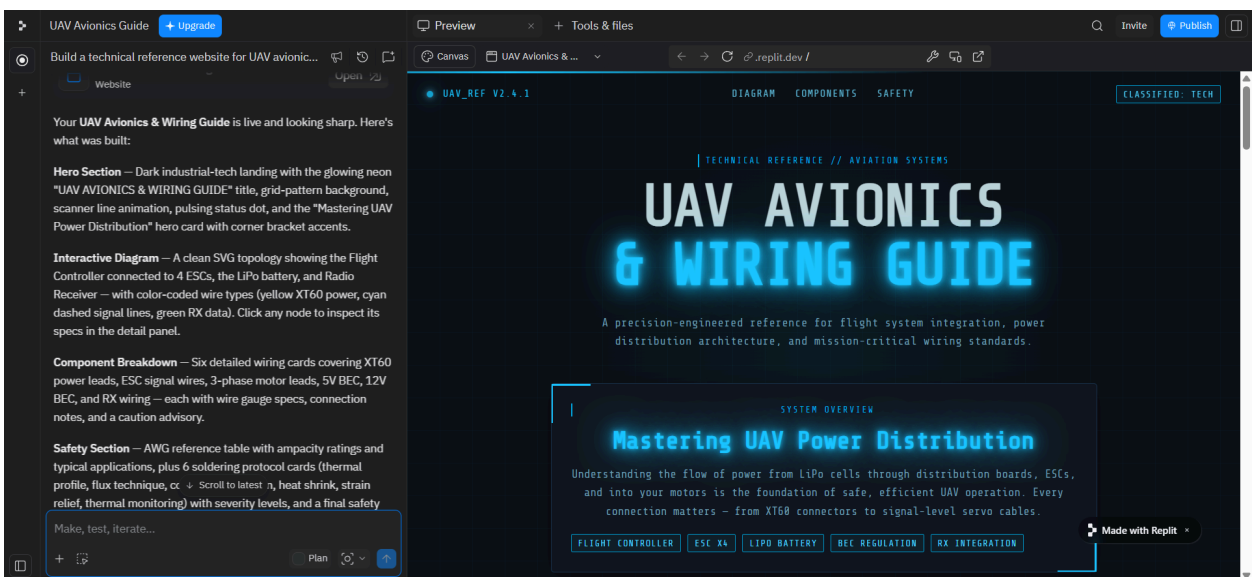

Figure 1.1. Initial Prompt in Replit

The goal of this laboratory is to develop a specialized one-page technical reference website titled UAV Avionics & Wiring Guide, specifically designed for aerospace engineering students and drone hobbyists. The site will feature an Industrial-Tech aesthetic, characterized by a high-contrast dark theme, neon blue accents, and a subtle grid-pattern background to mimic engineering graph paper. The layout begins with a high-impact Hero section labeled Mastering UAV Power Distribution, followed by the core feature: a centered, interactive SVG or CSS-styled diagram. This diagram visually maps the connections between a central Flight Controller (FC), four Electronic Speed Controllers (ESCs), a LiPo battery source, and a Radio

Receiver (RX). Complementing this visual is a detailed Component Breakdown that explains the wiring logic specifically focusing on signal wire routing, XT60 power leads, and the integration of 5V/12V BECs (Battery Eliminator Circuits). Finally, the page includes a critical safety section emphasizing the importance of proper wire gauge selection and soldering integrity to ensure flight stability and prevent electrical failure in Unmanned Aerial Vehicles.

Refinement Prompts

First Refinement Prompt:

"Enhance the wiring diagram. Make it so that when I hover over the 'Flight Controller' or 'ESC' boxes, a small tooltip appears explaining its function (e.g., 'FC: The brain of the drone'). Add color-coded lines for the wiring: Red for Power, Black for Ground, and Yellow for Signal. Use a monospace font like 'Courier New' for a more engineering-focused look."

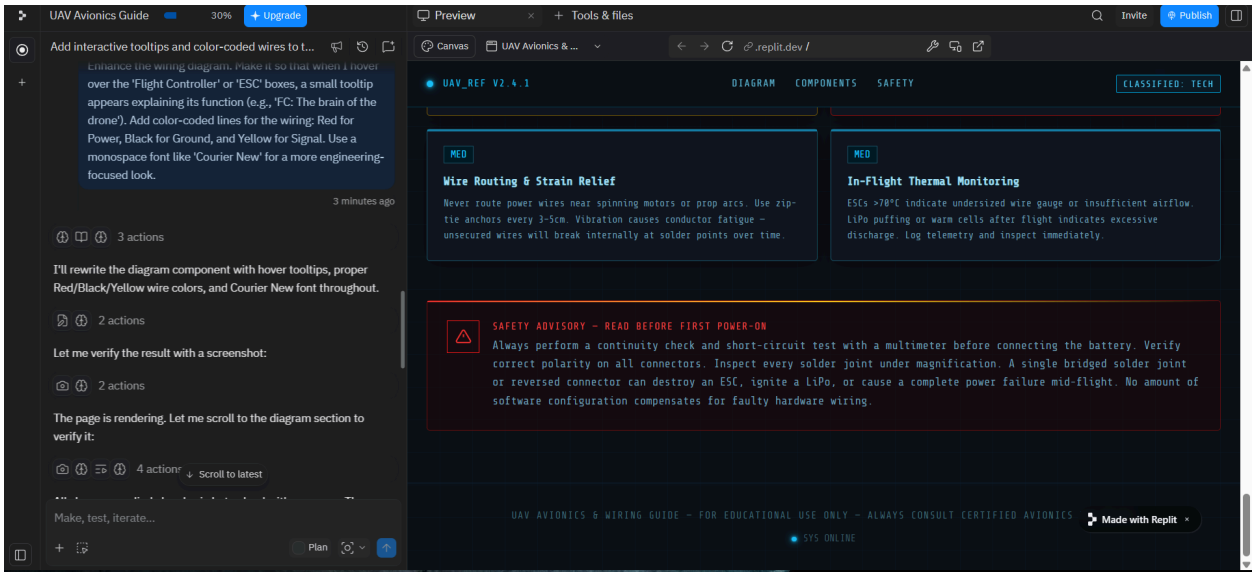

Figure 1.2. First Refinement Prompt in Replit

I instructed the AI to add hover-state tooltips to each component in the diagram to provide instant technical definitions for the FC, ESCs, and RX.

Second Refinement Prompt:

"Make the layout mobile-responsive so the diagram scales down properly on phones. Add a 'Safety Checklist' section at the bottom with checkboxes for 'Check for Short Circuits' and 'Verify Polarity.' Improve the button styling to have a glowing neon border effect when hovered."

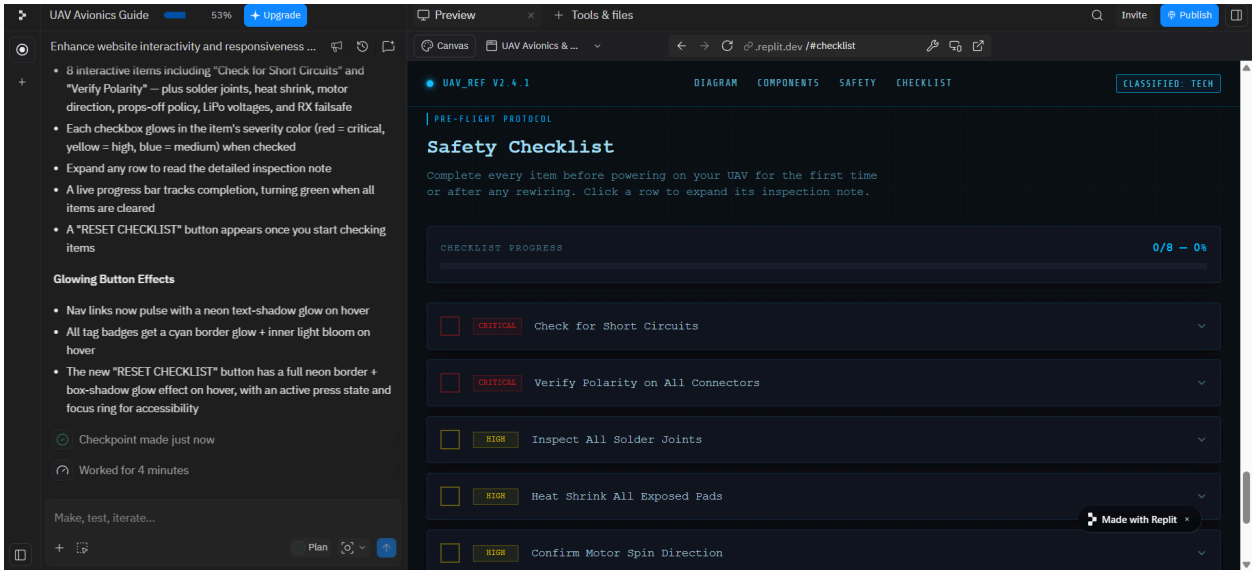

Figure 1.3. Second Refinement Prompt in Replit

I requested the addition of color-coded 'wires' (Red for VCC, Black for GND, and Yellow for PWM/Signal) to enhance the instructional clarity of the diagram, ensuring it met professional aerospace documentation standards.

Third Refinement Prompt:

"Add a 'Post-Wiring Safety Checklist' at the bottom with functional checkboxes for 'Continuity Check' and 'Polarity Verification.' Implement hover tooltips for the FC and ESCs that display their technical voltage ratings. Finally, add a 'Toggle Schematic' button that switches the diagram from a realistic layout to a simplified engineering symbol view for professional documentation."

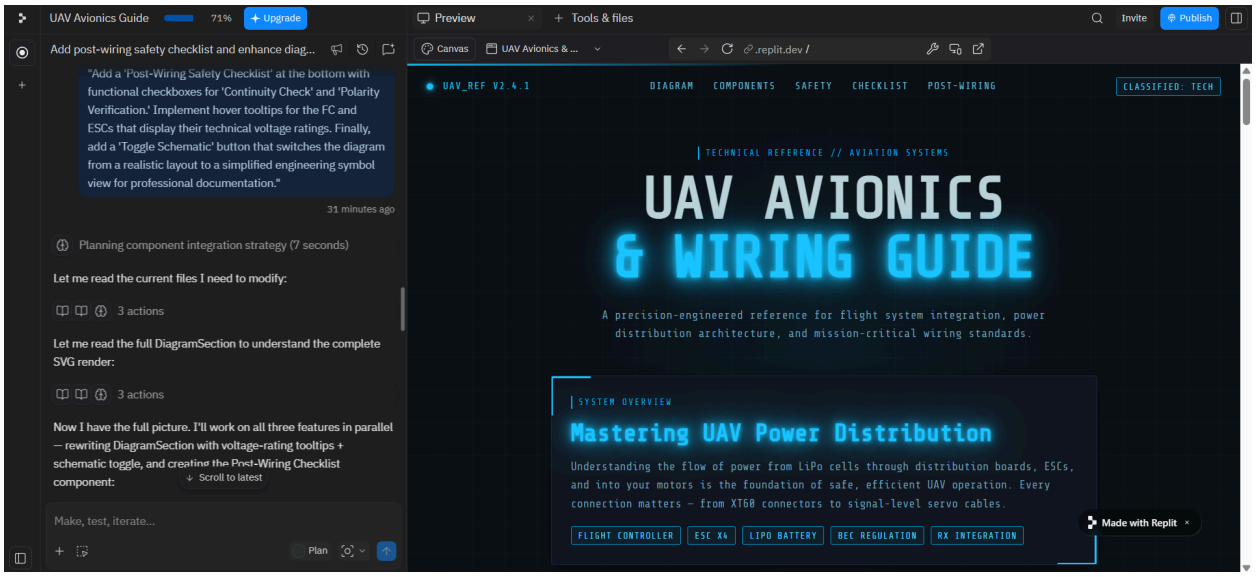

Figure 1.4. Third Refinement Prompt in Replit

This refinement was added to transition the project from a static visual guide into a functional pre-flight tool, ensuring the interface supports critical aerospace safety protocols and technical data verification.

Website Link

https://uav-avionics-guide--evidientespaulo.replit.app

Reflection

Replit was able to generate a structurally sound wiring diagram that aligned with professional engineering standards. The transition to an Industrial-Tech aesthetic was also seamless because the AI correctly interpreted the request for a grid-pattern background and neon blue accents to create a professional, schematic-like feel.

However, some manual fixing was required to refine the SVG pathing of the signal wires to ensure they did not overlap awkwardly. I also had to adjust the z-index of the hover tooltips to make sure they remained visible over the dark background elements. Through this process, I learned that iterative prompting is far more effective than trying to build a complex technical tool in a single step. By breaking the project into a foundation, a visual polish, and finally a functional utility layer, I was able to maintain better control over the code logic. I also discovered that providing the AI with a specific technical persona improved the relevance of the generated content and the accuracy of the component breakdowns.